



Mathematical Induction

PPT PRESENTED BY

NAME : Srinivas Rao Tammireddy

ROLL NO : 257Y1D5805

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What is Mathematical Induction?

Mathematical Induction is a proof technique used to establish that a statement $P(n)$ is true for ALL natural numbers $n \geq n_0$.

The Domino Analogy

Imagine an infinite row of dominoes labelled 1, 2, 3, ...

If (1) the FIRST domino falls, and (2) whenever a domino falls it knocks the NEXT one —
then ALL dominoes fall.

Mathematical Induction IS:

- ✓ A rigorous mathematical proof technique
- ✓ Proves statements for all $n \geq n_0$
- ✓ Uses logical deduction (deductive reasoning)
- ✓ 100% certain conclusion

Mathematical Induction is NOT:

- ✗ Inductive reasoning (science/empirical)
- ✗ "It works for first 100 cases, so always"
- ✗ A pattern guess or conjecture
- ✗ Trial and error

The Two Steps of Mathematical Induction

To prove $P(n)$ is true for all $n \geq 1$, we need exactly TWO steps:

Step 1: BASE CASE (The first domino falls)

Prove that $P(1)$ is true.

Directly verify the statement for the smallest value n_0 (usually $n_0 = 1$).

This is the 'anchor' — it grounds the proof in a concrete fact.

Step 2: INDUCTIVE STEP (Each domino knocks the next)

Inductive Hypothesis (IH): Assume $P(k)$ is true for some arbitrary $k \geq 1$.

Goal: Using the IH, prove that $P(k+1)$ is also true.

The IH is the key tool — we USE $P(k)$ as a 'known fact' to derive $P(k+1)$.

✓ **CONCLUSION:** If both steps succeed, then $P(n)$ is true for ALL $n \geq 1$ (by the Principle of Mathematical Induction)

Example: Setting Up the Proof

Theorem: $1 + 2 + 3 + \dots + n = n(n+1)/2$ for all integers $n \geq 1$

Step 1: Base Case ($n = 1$)

LHS (Left-Hand Side):

$$\text{Sum} = 1$$

RHS (Right-Hand Side):

$$1 \times (1+1) / 2 = 1 \times 2 / 2 = 1$$

LHS = RHS = 1 ✓ Base Case holds!

Step 2: Inductive Hypothesis

Assume $P(k)$ is true:

$$1 + 2 + 3 + \dots + k = k(k+1)/2$$

We call this the Inductive Hypothesis (IH).

We will USE this in the next step.

Goal of the Inductive Step: Prove $P(k+1)$

We need to show: $1 + 2 + \dots + k + (k+1) = (k+1)(k+2)/2$

Completing the Inductive Proof

We prove $P(k+1)$: $1 + 2 + \dots + k + (k+1) = (k+1)(k+2)/2$

Start: $1 + 2 + \dots + k + (k+1)$ *(The left-hand side of $P(k+1)$)*

Group the first k terms: $= [1 + 2 + \dots + k] + (k+1)$ *(Separate the last term)*

Apply the IH: $= k(k+1)/2 + (k+1)$ *(IH says the bracket $= k(k+1)/2$)*

Factor out (k+1): $= (k+1) \times [k/2 + 1]$ *(Common factor)*

Simplify: $= (k+1) \times (k+2)/2 = (k+1)(k+2)/2$ ✓ *(This is exactly $P(k+1)$!)*

✓ **CONCLUSION:** Both steps are complete. By the Principle of Mathematical Induction, $1+2+\dots+n = n(n+1)/2$ for ALL $n \geq 1$.

Strong Induction & Applications

Strong Induction (also called Complete Induction) assumes $P(j)$ is true for ALL $j \leq k$, not just $P(k)$. This stronger hypothesis is needed when $P(k+1)$ depends on multiple previous cases.

Strong Induction Steps

1. Base Case(s):

Prove $P(1)$ [and possibly $P(2)$, $P(3)$, etc.]

2. Strong IH:

Assume $P(j)$ is true for ALL $j \leq k$

3. Goal:

Prove $P(k+1)$ using any previous $P(j)$, $j \leq k$

Example: Fundamental Theorem of Arithmetic

Theorem:

Every integer $n > 1$ is either prime or can be written as a product of primes.

Why Strong Induction?

If n is composite, $n = a \times b$ where both $a, b < n$. We need induction applied to BOTH a and b .

Applications in CS:

Algorithm Correctness: Proving recursive algorithms (like Merge Sort, Binary Search) produce correct output for all inputs.

Complexity Proofs: Proving that algorithms run in $O(n \log n)$ time uses recurrences solved by induction.

Data Structure Props: Properties of heaps, trees, and balanced BSTs (height, node count) are proven by induction on height.

Formal Verification: Proving that program loops maintain invariants — induction on iteration count guarantees correctness.



THANK YOU

Department of Computer Science and Engineering
Marri Laxman Reddy Institute of Technology and Management